Department Of ISE/DS/IOT

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Problem Statement 1 : Load Monitoring Using IoT for GM University

1. Introduction

Electricity is a critical resource in any institution, including universities, where multiple buildings, laboratories, classrooms, and hostels consume significant energy. Efficient load monitoring and management are essential to optimize energy consumption, reduce electricity costs, and prevent power outages due to overload. Traditional energy monitoring methods rely on manual readings, which are time-consuming and prone to inaccuracies. With the advent of the Internet of Things (IoT), real-time load monitoring can be automated, providing precise insights into power consumption patterns.

2. Problem Statement

GM University faces challenges in monitoring and managing electricity usage across different departments. Unmonitored energy consumption leads to excessive electricity bills, inefficient power usage, and potential overloading of circuits. There is a need for a smart load monitoring system that can provide real-time data, alert users about anomalies, and optimize energy utilization. The proposed **IoT-based Load Monitoring System** will provide an intelligent solution by continuously tracking electricity consumption and helping administrators take data-driven decisions to enhance energy efficiency.

3. Objectives

The primary objectives of this project are:

- To develop an **IoT-based real-time load monitoring system** for GM University.
- To measure and analyze power consumption across different university buildings.
- To implement **automated alerts** for power surges, overloads, or abnormal consumption patterns.
- To enable **remote monitoring** of electrical loads via a web dashboard or mobile application.
- To generate **energy usage reports** and provide insights for better decision-making.
- To integrate predictive analytics to forecast energy consumption trends.

4. Technology Stack

The implementation of the **IoT-based Load Monitoring System** will involve the following technologies:

Hardware Components:

- **Microcontrollers**: ESP32 / Arduino with Wi-Fi capability
- **Current & Voltage Sensors**: ACS712 / PZEM-004T for power measurement
- **Wi-Fi Module**: Built-in ESP32 / External ESP8266
- **Relay Modules**: For controlling loads remotely
- **Cloud Storage**: Firebase / AWS IoT Core for data logging

Software & Cloud Services:

- **Embedded Programming**: Arduino IDE (C/C++) for microcontroller coding
- **Data Communication Protocols**: MQTT / HTTP for IoT communication
- **Cloud & Database**: Firebase / AWS DynamoDB for real-time data storage
- **Web Dashboard**: Node.js / Flask (Backend) + React.js / Angular.js (Frontend)
- **Mobile Application (Optional)**: Flutter / React Native for remote access
- **Data Visualization**: Grafana / ThingSpeak for analytics and reporting

Problem Statement 2 : Low-Cost Infotainment System for Buggies in GM University

1. Introduction

With the increasing adoption of **electric and solar-powered buggies** for campus transportation, universities are seeking ways to enhance the commuting experience for students, faculty, and visitors. A well-integrated infotainment system can provide real-time navigation, university announcements, event updates, and entertainment, making the ride more engaging and informative. However, traditional infotainment systems are often costly and require high-power consumption, which is unsuitable for energy-efficient buggies. A **low-cost, energy-efficient infotainment system** tailored for **solar-powered buggies** at GM University can address these challenges by offering a sustainable and user-friendly solution.

2. Problem Statement

GM University currently lacks an in-vehicle infotainment system for its campus buggies. This results in missed opportunities to **deliver campus-wide notifications**, promote events, guide visitors, and provide real-time transportation updates. Traditional infotainment systems available in commercial vehicles are expensive, require extensive wiring, and consume significant power. There is a need for a **cost-effective, lightweight, and solar-compatible infotainment system** that can operate efficiently on low power while offering interactive multimedia features for an enhanced commuting experience.

3. Objectives

The project aims to:

- Develop a **low-cost infotainment system** optimized for campus buggies.
- Integrate **GPS-based navigation** for route guidance within the university.
- Enable **real-time announcements and event updates** from the university administration.
- Provide **Wi-Fi / Bluetooth-enabled entertainment options** such as music streaming and educational content.

- Ensure **low power consumption and solar compatibility** for uninterrupted performance.
- Implement a **touchscreen or voice-assisted interface** for user-friendly interaction.
- Support future enhancements like **AI-based voice assistance** and emergency alert systems.

4. Technology Stack

Hardware Components:

- **Microcontroller**: Raspberry Pi 4 / ESP32 for infotainment processing
- **Display**: 7-10 inch LCD touchscreen / OLED panel
- **Speakers & Audio System**: Low-power stereo speakers with Bluetooth connectivity
- **Connectivity Modules**:
  - **Wi-Fi / Bluetooth** for media streaming and announcements
  - **GPS Module** for real-time navigation
- **Power Management**: Solar panel integration with battery backup

Software & Cloud Services:

- **Operating System**: Raspberry Pi OS / Custom Linux-based system
- **Media Player & Interface**: Python-based UI / Android-based infotainment UI
- **Cloud Services**: Firebase / AWS for real-time updates and announcements
- **Database**: SQLite / Firebase Firestore for storing announcements and navigation data
- **Networking**: MQTT / WebSocket for real-time updates
- **Mobile App (Optional)**: Flutter / React Native for user control and remote access
- **AI & Voice Control (Future Scope)**: Google Assistant / OpenAI Whisper for voice-based interaction

Problem Statement 3 : Accident Detection in Toilets and Camera-Restricted Areas

1. Introduction

Public and private spaces, including **toilets, restrooms, and other restricted areas**, often lack direct surveillance due to privacy concerns. However, these spaces are not exempt from potential hazards, such as **accidental falls, health emergencies, or slips**. Ensuring safety in such areas is critical, as the absence of surveillance cameras creates challenges in detecting incidents promptly, leading to **delayed medical response** and potential severe consequences. An **IoT-based accident detection system** can help **identify emergencies in real time** while respecting privacy.

2. Problem Statement

Currently, there is no **automated accident detection system** in **toilets, restrooms, and other camera-restricted areas**. In the event of an emergency, such as a person fainting, slipping, or suffering a medical condition (e.g., stroke or cardiac arrest), there is **no mechanism to alert authorities** in real-time. This delay can lead to life-threatening situations. Therefore, a **privacy-focused accident detection system** is needed to **monitor movement, detect unusual inactivity, and trigger alerts** without compromising personal privacy.

3. Objectives

The project aims to:

- Develop a **non-intrusive IoT-based system** to detect accidents in camera-restricted areas.
- Utilize **motion and environmental sensors** to detect falls, inactivity, or distress situations.
- Implement an **AI-based anomaly detection algorithm** to differentiate between normal and emergency scenarios.
- Enable **real-time alert notifications** to security personnel or medical teams.
- Integrate a **panic button system** for users to manually request assistance.
- Ensure **low power consumption** for continuous operation.
- Maintain **data privacy** and compliance with ethical monitoring practices.

4. Technology Stack

Hardware Components:

- **Microcontrollers**: ESP32 / Raspberry Pi for real-time processing
- **Sensors**:
  - **Passive Infrared (PIR) Sensors** – To detect movement and inactivity
  - **Ultrasonic Sensors** – To sense unexpected falls or unusual body positions
  - **Pressure Sensors** – Installed on the floor to detect sudden weight shifts
  - **Microphone (Optional)** – To recognize distress sounds (e.g., calls for help)
  - **Emergency Button** – For users to request assistance manually
- **Communication Modules**: Wi-Fi / LoRa / Bluetooth for real-time data transmission

Software & Cloud Services:

- **Embedded Programming**: C/C++ (Arduino IDE) / Python for sensor integration
- **Edge Computing**: AI-based anomaly detection using TensorFlow Lite
- **Data Processing & Analytics**: Node.js / Flask for backend processing
- **Cloud Storage & Alerts**: Firebase / AWS IoT for event logging and notifications
- **Mobile/Web Dashboard**:
  - React.js / Flutter for monitoring and alert response
  - Web-based admin panel for security and maintenance staff
- **Notification System**: SMS / WhatsApp API / Push notifications for emergency alerts

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